



Digital Watermarking for Deepfake Detection, Authentication, and Localization



Members: Jorge Luna, Joshua Rivera, Gian Miranda, Daniel Reyes, Revel Velazquez, Joshua Roman, Emmanuel Lopez

Advisors: Dr. Nayda Santiago, Dr. Alcibiades Bustillo

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1. Introduction

Manipulated and AI-generated media are increasingly realistic and difficult to distinguish from authentic media. This creates serious risks for misinformation, fraud, identity deception, and others forms of digital abuse. Digital watermarking can support media authentication by helping verify integrity, survive manipulations, and potentially localize edits. Deepfakes, especially facial manipulations, are the key motivating use case for this work.

2. Problem & Hypothesis

How effective is digital watermarking at preserving and verifying information in synthetically manipulated facial media?

Robustness:

How well does TrustMark survive common image transformations?

Face-Swap Resilience:

Can TrustMark remain recoverable after SimSwap face-swapping?

Localization:

Can watermark-based methods identify the spatial region of an image that has been manipulated?

3. Objectives

- Establish baseline watermark robustness under common image transformations.
- Evaluate watermark detectability and recoverability after AI-based face manipulation.
- Measure false-positive behavior in unwatermarked controls.
- Evaluate manipulation localization using predicted versus ground-truth masks.
- Identify limitations and promising directions for robust media authentication.

4. Methodology

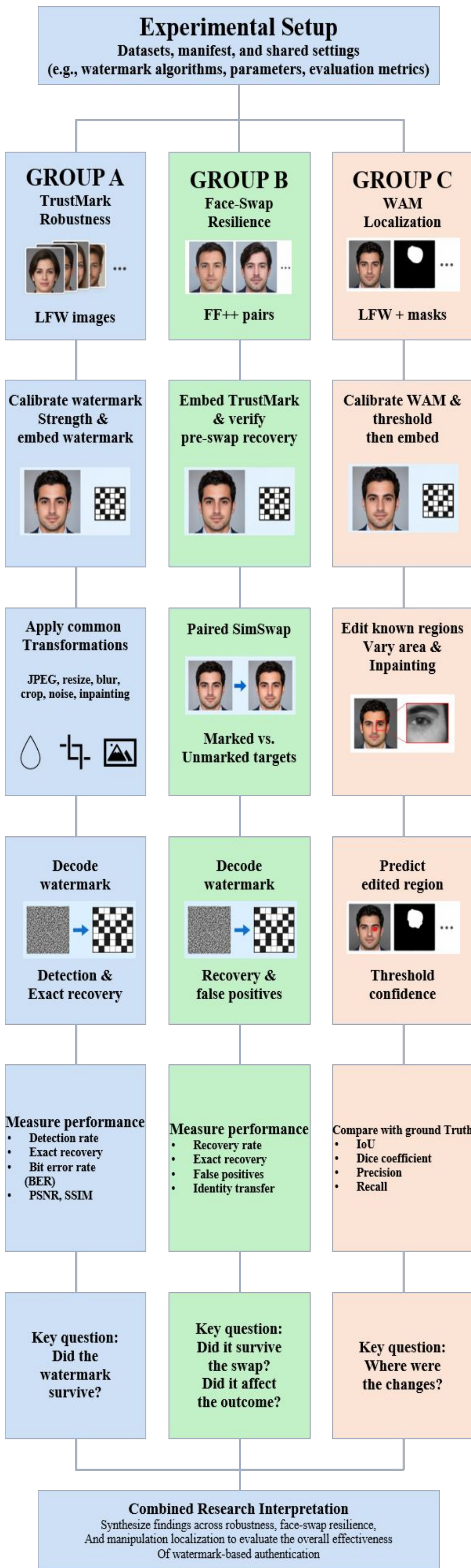


Figure 1: Three complementary experiments which evaluated watermark robustness, face-swap resilience, and manipulation localization, informed by recent watermarking and proactive forensic methods [1]-[3].

5. Results

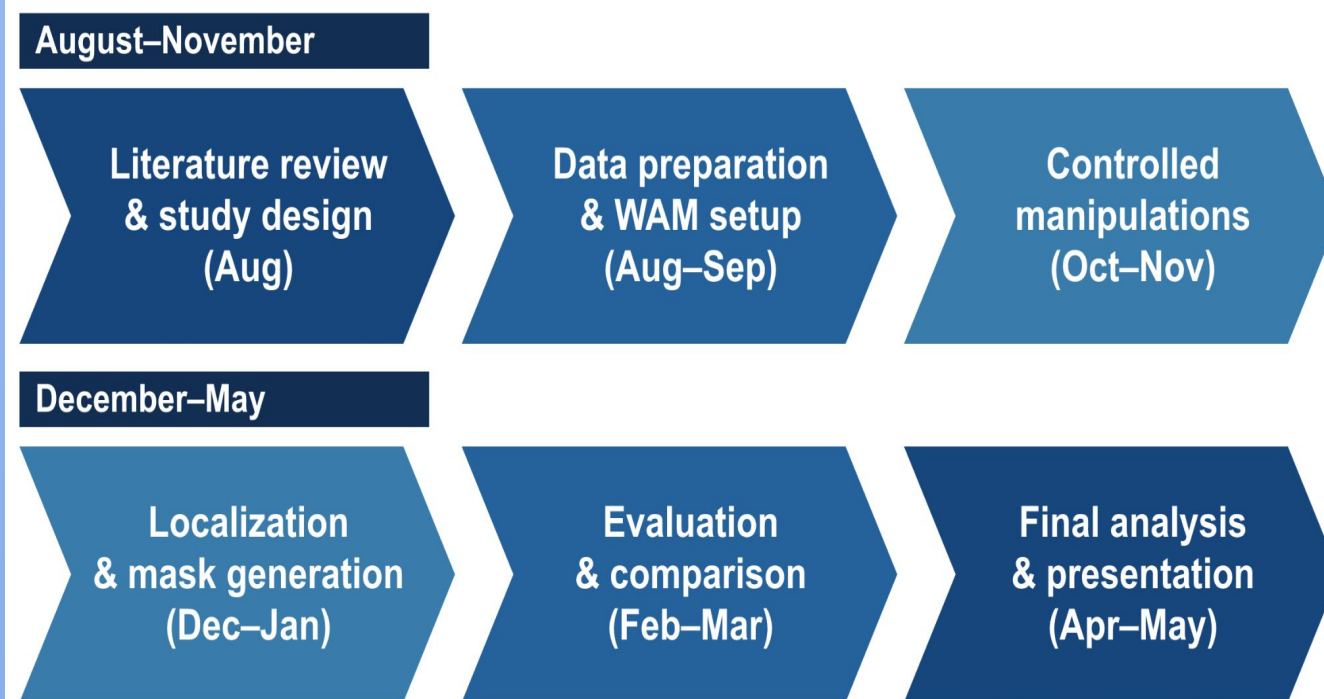
The three experiments evaluate complementary properties of watermark-based authentication: robustness to image transformations, resilience to face swapping, and localization of manipulated regions.

Group A - Watermark Robustness		
Metric	Condition	Result
Detection Rate	Clean	100%
Exact Recovery	Clean	100%
Detection Rate (Manipulated image)	Crop 0.5	2%
Exact Recovery (Manipulated image)	Crop 0.5	0%
Detection / Exact Recovery	Local facial manipulation	96%

Group B - Face Swap		
Metric	Before SimSwap	After SimSwap
Detection Rate	100%	100%
Exact Payload Recovery	100%	100%
Decoded Bit Accuracy	1.000	1.000
BER	0.000	0.000

Group C - Manipulation Location		
Method	Navier_Stokes	Telea
mean_iou	89%	90%
mean_dice	94%	95%
mean_precision	94%	95%
mean_recall	94%	94%

6. Timeline



References

- [1] T. Sander, P. Fernandez, A. Durmus, T. Furon, and M. Douze, “Watermark Anything with Localized Messages,” in Proc. Int. Conf. Learn. Representations (ICLR), 2025.
- [2] Bui, T., Agarwal, S., & Collomosse, J. (2025). TrustMark: Robust Watermarking and Watermark Removal for Arbitrary Resolution Images. Proceedings of ICCV 2025, 18629–18639.
- [3] Wu, X., Liao, X., Ou, B., Liu, Y., & Qin, Z. (2024). Are Watermarks Bugs for Deepfake Detectors? Rethinking Proactive Forensics. Proceedings of IJCAI 2024, 6089–6097.

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